

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 08

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = \text{£}11 \text{ million}$ (1)	1
01.2	<i>1 mark for each point.</i> • It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided (1) • ALARA: emissions should be As Low As Reasonably Achievable (1) • BATNEEC: the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this (1)	3
01.3	<i>1 mark for each point.</i> • As industries expand or cities grow the number of pollutant sources increases (1) • So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed (1)	2

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Critical Pathway Analysis: predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk; mainly radioactive discharges; checked by sampling • Critical Group Monitoring: exposure of the most-exposed group monitored before health effects; if their risk is acceptable everyone else is safer; emissions reduced if not • Emission control strategies: location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • Principles: polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • Selection of methods: production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • Evaluation: prevention usually cheaper and more effective than remediation; costs rise steeply near 100%; CPA relies on understanding of pathways; enforcement and international agreement needed for transboundary pollutants; judgement about acceptable risk is political as well as scientific • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
02.1	<i>Correct answer only.</i> • C (1)	1
02.2	<i>1 mark for each point.</i> • Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms (1) • Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas (1)	2

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> • HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere (1) • So most do not persist long enough to reach the stratosphere where UV would release their chlorine (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
03.1	<i>1 mark for each point.</i> • Terracing: sloping land divided into narrow level fields held by retaining walls along contours; water slows as it crosses each terrace and infiltrates instead of running off (1) • Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff (1) • Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles; protect bare fields after harvest (1) • Mulching: organic matter on the surface protects soil from raindrop impact and wind; decomposes to humus which binds particles; retains moisture (1) • Strip cropping: strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field; slows runoff (1)	5
03.2	<i>1 mark for each point.</i> • No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles (1) • Permanent root systems and canopy cover protect the soil all year (1)	2
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Control stocking density so vegetation regrows faster than it is eaten (1) • Restrict access to vulnerable areas such as river banks (fencing) (1) • Move / remove livestock at high-risk times, eg after heavy rain when soil is wet (1) • Rotational grazing to rest pastures (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	
04.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(7.7 - 3.9) / 7.7 \times 100$ (1) • = 49.4% (accept 49%) (1)	2
04.2	<i>1 mark for each point.</i> • Floating sea ice has already displaced its own mass of water; when it melts it contracts and occupies the same volume as the ice below the water line (1) • Melting land ice (glaciers, ice sheets) adds new water to the sea, increasing its volume (1)	2
04.3	<i>1 mark for each point.</i> • Ice and snow have a high albedo and reflect most sunlight; open sea water has a low albedo (1) • As ice area shrinks more sunlight is absorbed by the sea and converted to heat, raising temperatures (1) • This melts more ice: a positive feedback mechanism; thinner ice melts more easily each summer (1)	3

Question	Marking guidance	Marks
04.4	<i>1 mark for each point.</i> - Thermal expansion: warmer sea water expands (slow because of the huge volume and high heat capacity; only surface water warms directly) (1) - Melting of land ice (glaciers, Greenland and Antarctic ice sheets) adds water to the oceans (1)	2
04.5	<i>1 mark.</i> Ice shelves grounded on the seabed hold back / block the forward movement of the glaciers or ice sheet behind them; once removed the land ice flows faster (eg West Antarctica) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
05.1	<i>Correct answer only.</i> Aerobic decomposition produces little methane and returns nutrients to the soil (1)	1
05.2	<i>1 mark for each point.</i> - It may contain concentrated heavy metals / toxic residues (1) - It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble (1)	2
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Impermeable polymer / clay liner to stop leachate escaping (1) - Collection and treatment of leachate (1) - Collection of methane and use as fuel (1) - Daily covering of waste / final capping and restoration (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
06.1	<i>1 mark for each point.</i> - Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening (1) - Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight (1) - NO₂ (from morning traffic) is broken down by UV to NO and monatomic oxygen; monatomic oxygen reacts with O₂ to form ozone (1) - Ozone lags behind the NO / NO₂ peaks (which are produced by the morning rush-hour traffic) because the reactions take time; falls at night when there is no sunlight (1)	4
06.2	<i>1 mark for each point.</i> - NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure (1) - NO is oxidised in the atmosphere (by oxygen / ozone) to NO₂, a secondary pollutant (1)	2
06.3	<i>1 mark for each point.</i> - PANs (peroxyacetyl nitrates) (1) - Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NO_x) (1)	2
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Heavy vehicle use giving high NO_x and hydrocarbon emissions (1) - Sunny climate providing the UV for photochemical reactions (1) - Valley / basin topography and frequent temperature inversions trapping pollutants (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
07.1	<i>1 mark for each point.</i> - Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports (1) - Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones; open-cast mines take far more land than deep mines (1)	2
07.2	<i>1 mark for each point.</i> - Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle (1) - Noise: baffle mounds / embankments absorb and deflect; blasting at set, predictable daytime hours; quieter machinery (1) - Turbid water: sedimentation lagoons where water stands until solids settle before discharge (1) - Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals (1) - Amenity: landscaping and tree planting to screen the mine; restoration plans agreed as a condition of planning permission (1)	5
07.3	<i>1 mark for each point.</i> - Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface (1) - At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village (1) - Killing 144 people, mainly children in the primary school (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	

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08.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Uses / benefits: nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • Risks: free radical production, DNA damage, mutation, cancer; acute and chronic, somatic and gonadic effects; contamination vs exposure; activation products; long-lived wastes; accidents (Chernobyl, Fukushima); weapons fallout • Factors affecting risk: type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • Principles: ALARA, BATNEEC; closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • Monitoring: dosimeters, film badges, air and contamination monitors; environmental monitoring via CPA and CGM; waste categories and disposal (vitrification, encapsulation, Drigg) • Evaluation / judgement: benefits generally outweigh risks when controls are applied; public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9
08.2	<i>1 mark.</i> • Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	
09.1	<i>Correct answer only.</i> • B - rivers are accessible over a large area with few natural contaminants (1)	1

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)	2
09.3	<i>1 mark for each point.</i> • Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
10.1	<i>1 mark for each point.</i> • t-test (or Mann-Whitney U if not normally distributed) (1) • Comparing two means of counted / measured data from two independent samples (1)	2
10.2	<i>1 mark for each point.</i> • The difference is statistically significant (less than 1% probability that it arose by chance) (1) • The null hypothesis (no difference between fields) is rejected; the organic field has significantly more earthworms (1)	2
10.3	<i>1 mark for each point.</i> • A significant difference does not show a causal relationship (1) • Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference (1)	2

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10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Scientific methodology: identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • Sample location: random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias; interval chosen so variation is detected without excess data • Sample timing: diurnal, seasonal, weather-related and long-term changes; sampling on different occasions; interval matched to rate of change • Sample size and number: larger and more samples for non-homogeneous variables; running mean and standard deviation show when the mean is reliable; preliminary study to decide • Standardisation and control of variables; accuracy, precision, reliability, representativeness; anomalies and repeats; calibration of equipment; appropriate techniques and risk management • Statistical analysis: choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • Evaluation: trade-offs between cost, time, disturbance and reliability; limitations of sub-sampling; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	

Question	Marking guidance	Marks
11.1	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25